KIET Deemed to be University, Ghaziabad

Assignment: Data Flow Diagram - Level 0 (Context Diagram)

Name: Ashutosh Tiwari | Branch: CSE | Section: B | Semester: V

University Reg. No.: 202401100200098 | University Roll No.: 2400290100098 | Project: Digital Payment Gateway Simulation

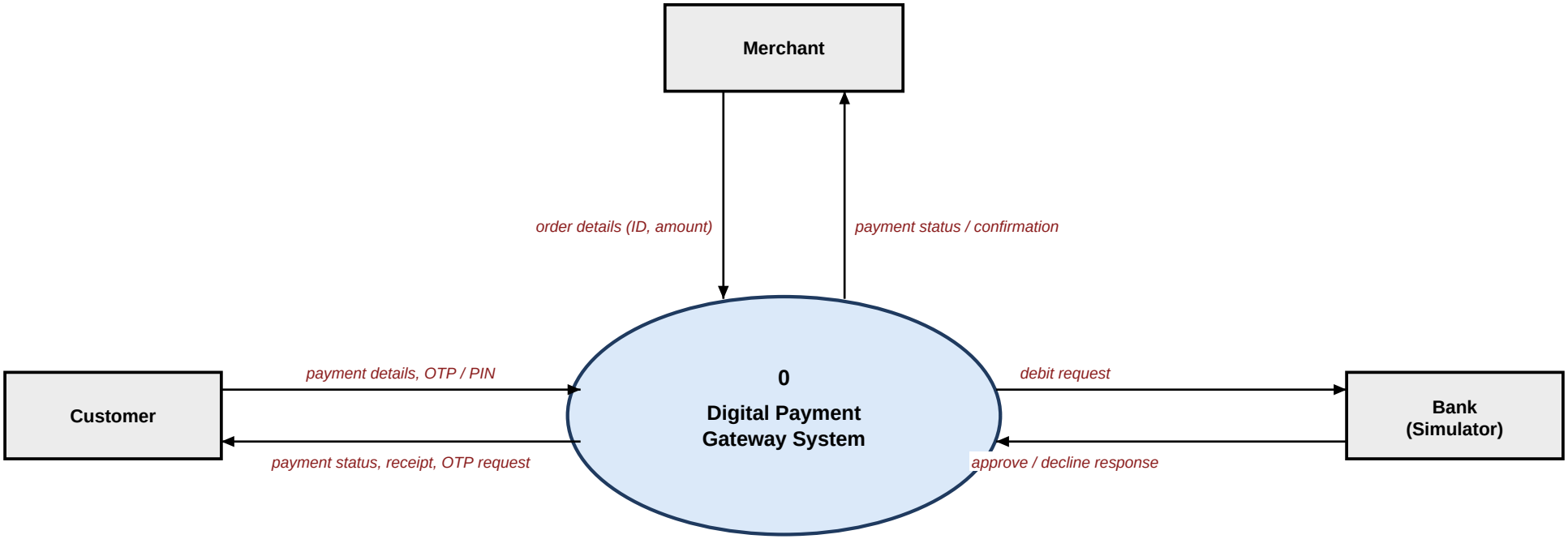

Fig. 1: Level 0 DFD (Context Diagram) of the Digital Payment Gateway Simulation

KIET Deemed to be University, Ghaziabad

Assignment: Data Flow Diagram - Level 1

Name: Ashutosh Tiwari | Branch: CSE | Section: B | Semester: V

University Reg. No.: 202401100200098 | University Roll No.: 2400290100098 | Project: Digital Payment Gateway Simulation

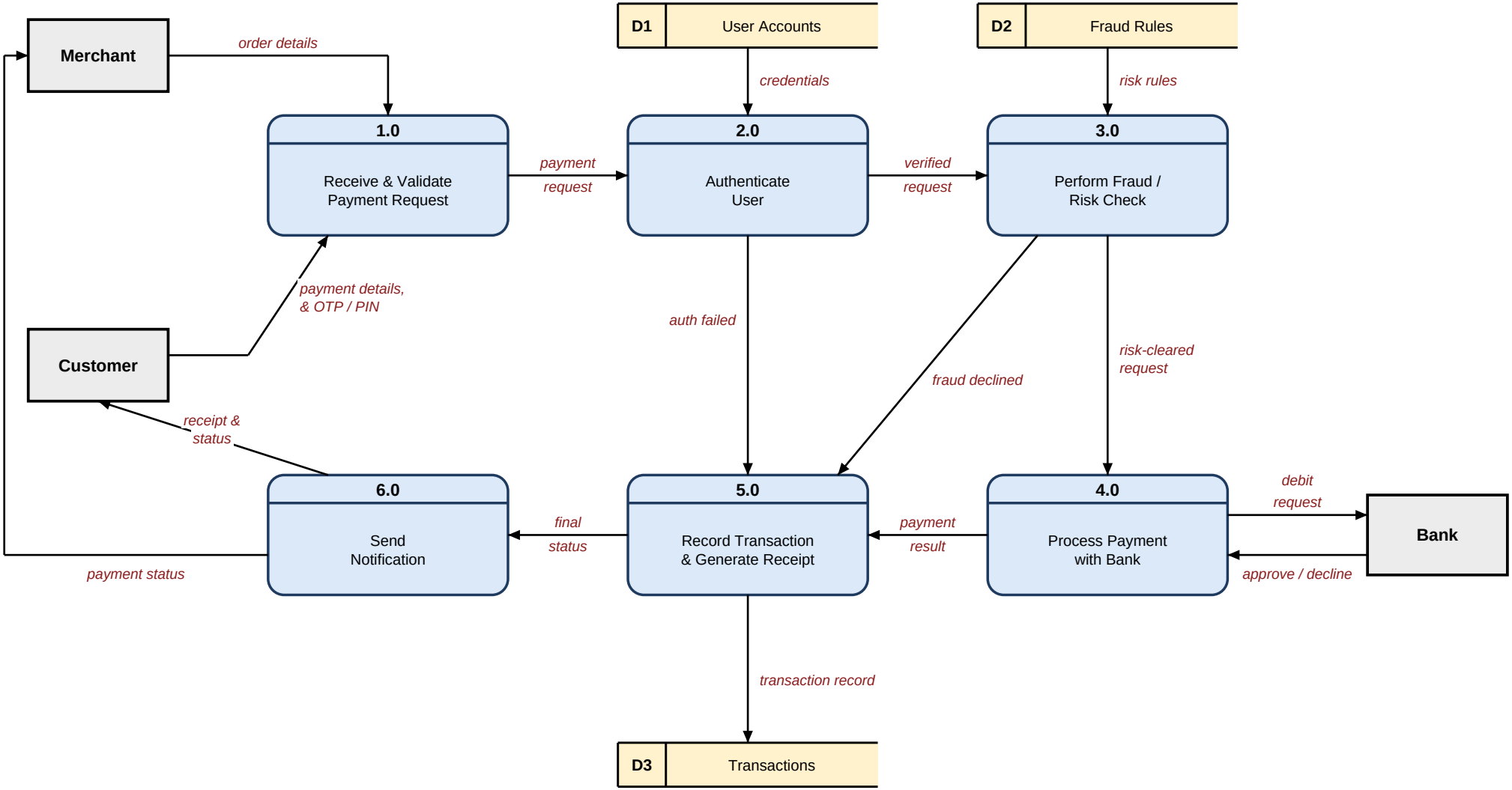

Fig. 2: Level 1 DFD of the Digital Payment Gateway Simulation (Gane-Sarson notation)

Rounded box = process Grey box = external entity Open box (D1-D3) = data store Arrow = data flow

KIET Deemed to be University, Ghaziabad

Assignment: Data Flow Diagram - Description

Name: Ashutosh Tiwari Branch: CSE Section: B Semester: V
University Reg. No.: 202401100200098 University Roll No.: 2400290100098 Project: Digital Payment Gateway Simulation

Type	Name	Description / Data handled
External entity	Customer	Gives payment details and OTP/PIN; receives receipt and payment status
External entity	Merchant	Sends order details (order ID, amount); receives payment status
External entity	Bank (Simulator)	Receives debit request; returns approve / decline response
Process	1.0 Receive & Validate Payment Request	Checks that order and payment details are complete and valid
Process	2.0 Authenticate User	Verifies credentials and OTP/PIN using stored user data
Process	3.0 Perform Fraud / Risk Check	Applies fraud rules (amount limit, repeated attempts)
Process	4.0 Process Payment with Bank	Sends debit request to the bank and receives the result
Process	5.0 Record Transaction & Generate Receipt	Stores the transaction (SUCCESS / FAILED) and creates the receipt
Process	6.0 Send Notification	Sends SMS / e-mail and returns status to the merchant and customer
Data store	D1 User Accounts	User and merchant credentials, linked accounts
Data store	D2 Fraud Rules	Amount limits and risk rules used in the risk check
Data store	D3 Transactions	Transaction ID, amount, status, time stamp, payment method
Data flows	Failure flows	'auth failed' (2.0 to 5.0) and 'fraud declined' (3.0 to 5.0) record the failed transaction